

Rank Before You Merge: Training-Free Machine-Centric Visual Token Compression for Efficient Vision-Language Model Inference

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Abstract

Native spatial mergers already compress visual patches before a vision-language model (VLM) reaches its language model, yet most token selectors rank the merger outputs. We call compression *machine-centric* when its units, budget, and interface follow the deployed model’s native computation graph. Rank-Before-Merge (RBM) is a training-free rule that ranks complete native units before applying the unmodified merger. At 25% retention, RBM outperforms matched post-merger L2 selection on all nine text-dense full-split cells across Qwen3-VL, Qwen2.5-VL, and InternVL3, with gains of 11.0–38.4 percentage points and 235–432 OCRBench points. A merger-aware rank prior also improves FastV-k3 on all four locked Qwen3-VL probes at the same budget, although only TextVQA is significant and pure RBM remains stronger on OCRBench. Rank and token-survival analyses attribute the main effect to merger-induced saliency rewriting. RBM adds no learned parameters and increases measured throughput by 68% on Qwen3-VL at 25% retention.

Introduction

Modern VLMs can expose hundreds to tens of thousands of visual tokens to the language model. Reducing that sequence lowers attention and key–value-cache cost, so visual-token compression has become an important part of efficient multimodal inference. On our A40 system, reducing the visual-token rate to 25% lowers Qwen3-VL latency from 0.36 to 0.23 seconds and raises throughput by 68%. The central question is which visual evidence should occupy that limited rate.

Merger-equipped models introduce a structural detail that current methods often treat as fixed plumbing: neighboring patch features are first compressed into native spatial units, and a later selector decides which merged tokens survive. If the merger changes saliency order, a precise top- k decision made after it cannot recover an answer-bearing unit that was ranked below the threshold.

We call a compressor *machine-centric* when its selection unit, budget, position handling, stream identity, and output interface follow the native computation graph of the deployed model. This term does not imply a new learned importance model. It distinguishes compression that respects the machine’s existing representation and topology from a selector applied to a convenient but already altered token stream. Figure 1 shows the resulting design: Rank-Before-Merge (RBM) evaluates complete 2×2 units on merger-input features, keeps a fixed per-image budget, and invokes the original merger only on the survivors.

Our first contribution is a representation-dependent formulation of visual token selection around a lossy native merger. It states exactly when pre- and post-merger

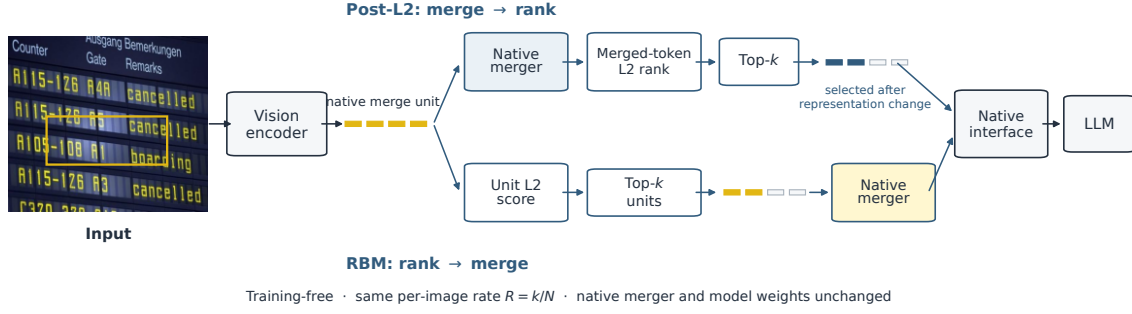


Figure 1: RBM follows the native merge topology. It ranks complete merger-input units, preserves a fixed per-image budget, and passes the selected units through the unmodified merger and language-model interface. The comparison changes selection order, not model weights.

top- k decisions agree. Second, we introduce RBM as a training-free compressor with an explicit native-unit and interface contract. Across three VLMs and two merger designs, RBM preserves substantially more text-dense evidence than matched post-merger selection at equal token count. Third, we test whether the pre-merger signal complements a strong query-conditioned selector. A locked RankBridge probe reserves 20% of the final budget for pre-merger-ranked units and fills the remaining budget using FastV attention. It improves FastV-k3 on all four tested Qwen3-VL benchmarks by 0.5–3.2 percentage points, with a significant TextVQA gain. This establishes that pre-merger and query-conditioned ranks contain complementary signals. Finally, controlled rank, edge-energy, and ranking-swap analyses connect the accuracy gap to merger-induced saliency rewriting.

Related Work

Existing methods intervene at different points in the inference graph. FastV uses question-conditioned attention inside an early LLM layer [1]; PyramidDrop progressively shortens the sequence across LLM layers [2]; FasterVLM and SparseVLM refine attention- and query-based selection [3, 4]. VisionZip selects dominant visual tokens and merges the remainder, while AdaptMerge studies adaptive visual–language-guided merging [5, 6]. PruMerge, FitPrune, and recent systems add learned, fitted, diversity, or text-guided importance signals [7, 8, 9, 10, 11].

These methods chiefly optimize *which* token survives. We isolate *where in the representation path* the decision is made. The distinction is material for Qwen-VL: its PatchMerger concatenates four neighboring patch vectors, then applies LayerNorm and a nonlinear MLP. InternVL3 instead uses pixel-shuffle downsampling [12, 13, 14]. RBM keeps these native operators and the downstream visual interface unchanged; it moves the ranking decision to their input.

Machine-Centric Compression

Selection around a native merger

Let $U = \{u_i\}_{i=1}^N$ be the native merge units and let M be the model’s per-unit merger. A representation-dependent scorer produces $a_i = s_{\text{pre}}(u_i)$ before merging and $b_i = s_{\text{post}}(M(u_i))$ after merging. At budget k , the two decisions are

$$K_{\text{pre}} = \text{TopK}_k\{a_i\}, \quad K_{\text{post}} = \text{TopK}_k\{b_i\}. \quad (1)$$

With deterministic tie breaking, they are equivalent exactly when $K_{\text{pre}} = K_{\text{post}}$. A sufficient condition for agreement at every budget is $b_i = g(a_i)$ for a strictly increasing g . We measure the violation at budget k by

$$D_k = 1 - \frac{|K_{\text{pre}} \cap K_{\text{post}}|}{|K_{\text{pre}} \cup K_{\text{post}}|}. \quad (2)$$

For a fixed unit set, selection and a per-unit merger are compatible. The non-equivalence arises because the merger changes the ranking that determines the set.

For compression analysis, define the visual-token rate and task distortion as

$$R = \frac{k}{N}, \quad \mathcal{D}(R) = 1 - \frac{S(R)}{S(1)}, \quad (3)$$

where $S(R)$ is the downstream benchmark score at rate R . A lower $\mathcal{D}$ means that more task-relevant information survives. RBM and Post-L2 have identical R , so their vertical separation isolates distortion introduced by the representation used for selection.

Rank-Before-Merge

For every image, RBM groups ViT patch features into the model’s native 2×2 units. For a unit u with input patch vectors $h_{u,1:4}$, it uses the query-blind control score

$$s_{\text{pre}}(u) = \frac{1}{4} \sum_{j=1}^4 \|h_{u,j}\|_2, \quad s_{\text{post}}(u) = \|M(h_{u,1:4})\|_2. \quad (4)$$

It retains $k = \max(1, \text{round}(\kappa N))$ units and applies the native merger to those units. The method contract has four parts: selection uses complete native units; every image receives its exact top- k budget; the same unit identities propagate through main and deepstack streams; and the native merger, positional encoding, visual interface, and downstream LLM are unchanged. L2 scoring costs $O(Nd)$ for d -dimensional features and top- k selection costs $O(N \log k)$ with a heap. RBM adds no parameters and does not require question tokens or an LLM-attention pass.

Because Qwen’s merger is applied independently per unit, supplying a fixed kept set to the original operator produces the same merged representations. This is byte-identical in the verified Qwen3-VL path. The invariant lets us separate a changed ranking from a changed forward operator in the swap control below.

Kept visual units (blue) per selector — two OCR cases

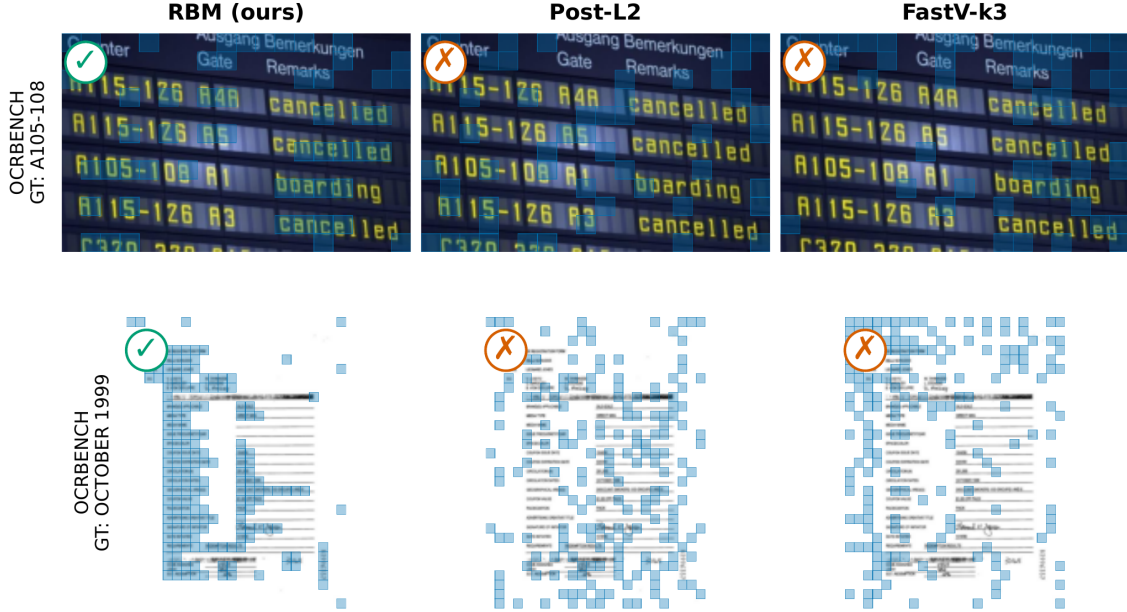


Figure 2: Audited OCRBench examples at 25% retention. RBM, Post-L2, and FastV-k3 preserve different spatial evidence. The examples illustrate selection behavior; aggregate results are reported separately.

A merger-aware prior for FastV

FastV ranks the full visual-token candidate set with question-conditioned attention at LLM layer $K = 3$. RankBridge tests whether the merger-input order contains information that this later signal misses. At retention $\kappa = 0.25$, it protects $q = \text{round}(\rho k)$ positions selected by the RBM pre-rank and fills the remaining $k - q$ positions by FastV rank. A development gate over $\rho \in \{0.1, 0.2, 0.3\}$ selected $\rho = 0.2$; the value was then locked for four $n = 200$ benchmarks. At $\rho = 0$, the implementation is bit-identical to FastV; at $\rho = 1$, its kept set is identical to RBM. All visual tokens remain available until layer K , so the pre-rank does not starve FastV’s candidate set.

Experimental Protocol

We evaluate Qwen3-VL-8B, Qwen2.5-VL-7B, and InternVL3-8B. The Qwen models use a learned PatchMerger and InternVL3 uses pixel-shuffle downsampling. All runs use bf16 weights, greedy decoding, and family-correct native positions. RBM, no-compression, and Post-L2 cells use vLLM 0.19 V1 [15]; FastV and RankBridge use a HuggingFace eager-attention harness because FastV has no vLLM path. The eager implementation reproduces the native path on 8/8 per-sample checks, and HF/vLLM no-compression outputs agree on 16/16 held-out cases. Throughput comparisons remain within vLLM.

The benchmarks are TextVQA with official VQA accuracy [16], DocVQA with ANLS [17], OCRBench with its /1000 score [18], and GQA with normalized exact

Table 1: Official full-split results. RBM and Post-L2 use the same per-image visual-token count. Δ is RBM minus Post-L2 at 25% retention. OCRBench uses points; other rows use percentage points.

Model	Benchmark (metric)	None	RBM	Post-L2	RBM-Post
Qwen3-VL	TextVQA (VQA acc.)	.844	.605	.222	+38.4
	DocVQA (ANLS)	.956	.481	.238	+24.3
	OCRBench (/1000)	760	547	184	+363
	GQA (EM)	.616	.449	.477	-2.8
Qwen2.5-VL	TextVQA (VQA acc.)	.862	.702	.442	+26.1
	DocVQA (ANLS)	.949	.636	.526	+11.0
	OCRBench (/1000)	817	480	182	+298
	GQA (EM)	.604	.559	.585	-2.6
InternVL3	TextVQA (VQA acc.)	.8338	.7890	.4148	+37.4
	DocVQA (ANLS)	.9221	.7284	.3820	+34.6
	OCRBench (/1000)	852	753	321	+432
	GQA (EM)	.6293	.5993	.6031	-0.4

match [19]. Full splits contain 5000, 5349, 1000, and 12578 items. Scorers are ports of the lmms-eval implementations [20]. Pre/post comparisons pair identical samples and per-image token counts. The nine prespecified text-dense comparisons at 25% retention use 20,000-resample paired-bootstrap intervals and paired sign-flip tests with Holm correction; every corrected p is at most 4.5×10^{-4} . FastV and RankBridge comparisons are secondary locked probes and use paired tests on $n = 200$.

Compression Results

Matched pre- and post-merger selection

Table 1 isolates selection stage within the same query-blind L2 family. At 25% retention, RBM outperforms Post-L2 in all nine text-dense cells. The Qwen3-VL gains are 38.4 pp on TextVQA, 24.3 pp on DocVQA, and 363 OCRBench points; Qwen2.5-VL gains 26.1 pp, 11.0 pp, and 298 points. The direction replicates with InternVL3’s different merger: 37.4 pp, 34.6 pp, and 432 points. Deeper 12.5% retention widens the gap on the reported text-dense cells. RBM is not lossless relative to no compression, especially on InternVL3 DocVQA and OCRBench. The evidence establishes improved retention over a matched post-merger decision.

GQA gives the boundary of the result. The operational Qwen rankings favor Post-L2 by 2.6–2.8 pp, while InternVL3 differs by only -0.4 pp with a 95% interval crossing zero. A full-split Qwen3-VL matched-boundary control ranks the pre arm at the main merger’s own input. It confirms before-merger gains of 27.68 pp on TextVQA, 4.59 pp on DocVQA, and 235 OCRBench points, but favors Post-L2 by 5.64 pp on GQA (95% CI $[-6.42, -4.86]$). The result is a workload-conditioned stage effect, not a universal preference for early selection.

Table 2: Locked Qwen3-VL probe at 25% retention ($n = 200$). RankBridge protects 20% of its final budget with the pre-merger rank and fills the remainder with FastV-k3 attention.

Benchmark	RBM	FastV-k3	RankBridge	RB–FastV (pp)
TextVQA	.5967	.7633	.7950	+3.2
DocVQA	.4239	.5863	.6023	+1.6
OCRBench	.5801	.4586	.4641	+0.6
GQA	.4150	.5050	.5100	+0.5

FastV and the merger-aware prior

Table 2 reports the locked RankBridge probe at the same 25% budget. The merger-aware prior improves FastV-k3 on every benchmark. TextVQA gains 3.2 pp with paired $z = 2.11$; the 0.5–1.6 pp gains on the other three benchmarks are not significant at $n = 200$. Pure RBM remains 11.6 pp stronger than RankBridge on OCRBench, so the hybrid fails a preregistered requirement to stay within 1 pp of the stronger parent on every benchmark. This negative gate is informative: pre-merger rank contributes information beyond query attention, but a single global mixture cannot replace the workload boundary.

The separate same-scope baseline campaign also shows why the pure methods remain useful. On full OCRBench, RBM exceeds FastV by 14.6 pp on Qwen3-VL and 8.7 pp on Qwen2.5-VL. FastV is stronger on Qwen3-VL TextVQA and GQA and on Qwen2.5-VL TextVQA. Query-conditioned selection and topology-preserving selection therefore occupy different regimes.

Figure 3 expresses the same budget sweep as task rate–distortion operating points. At equal rate, the pre-merger decision sharply lowers distortion on text-dense tasks. GQA reverses the ordering, making the workload boundary visible in the same compression coordinates.

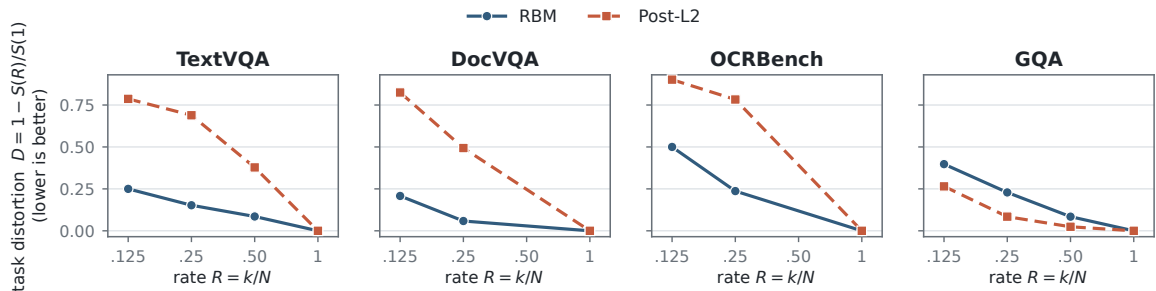


Figure 3: Task rate–distortion operating points on the locked Qwen3-VL diagnostic subset ($n = 200$ per point). Rate is the retained native visual-unit fraction; distortion is the relative loss in official task score from the uncompressed model. Curves connect measured rates only.

Why the Merger Changes Selection

Figure 4 summarizes three diagnostics on Qwen3-VL. First, pre- and post-merger unit ranks are weakly related. Spearman correlation is 0.137 on DocVQA, 0.332 on TextVQA, and 0.360 on GQA; top-25% kept-set Jaccard is 0.180, 0.243, and 0.278. The native merger therefore changes the actual top- k decision, not only score scale.

Second, the changed order is structured. With rank shift defined as post rank minus pre rank, its correlation with Sobel edge energy is +0.439 on DocVQA, +0.155 on TextVQA, and +0.036 on GQA. On DocVQA, units kept by RBM but dropped by Post-L2 have mean edge energy 0.641, compared with 0.124 in the reverse group; 92% lie above the per-image median, compared with 35%. This supports text-stroke demotion as a mechanism-level association. It does not establish the merger’s training objective as the cause.

Third, a ranking-swap control holds the post-merger forward path fixed but uses the pre ranking. On Qwen3-VL, it selects exactly the RBM set (Jaccard = 1.000) and recovers pre accuracy, while the retained merged outputs are byte-identical. The measured Qwen3-VL gap is therefore attributable to the selection ranking rather than a different operation on the kept units.

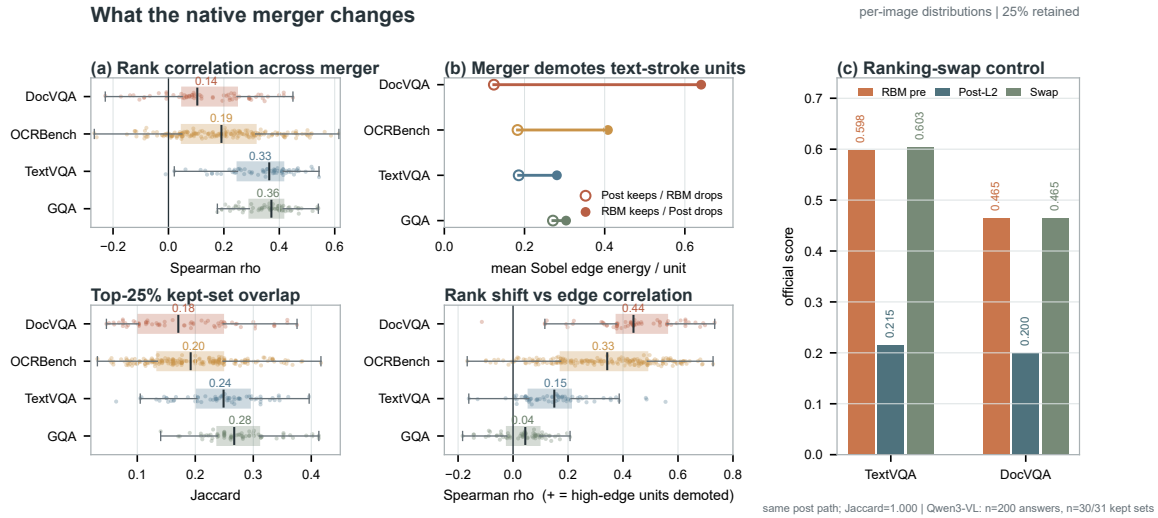


Figure 4: The native merger rewrites selection saliency. Rank agreement is low, text-stroke-rich units are demoted, and a ranking-swap control restores RBM accuracy with the same kept set.

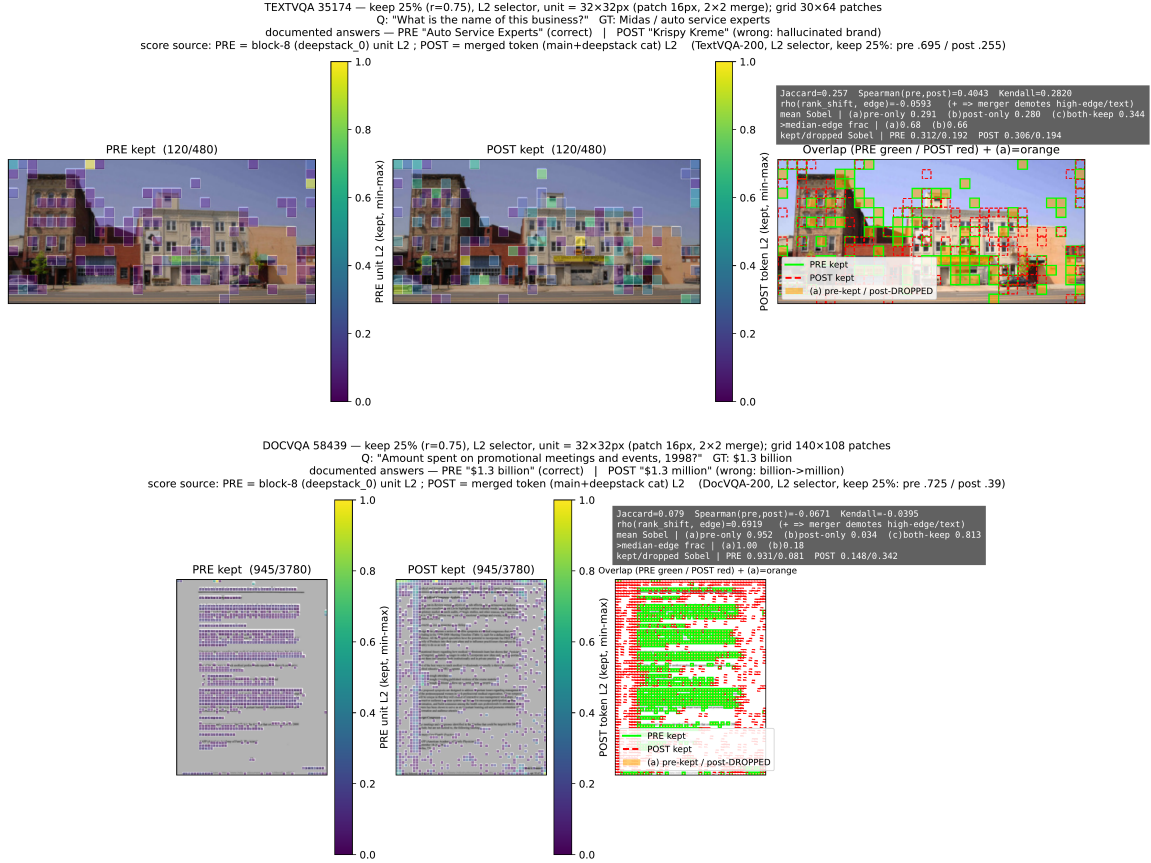


Figure 5: Token-survival maps for audited TextVQA and DocVQA images. Each row shows RBM, Post-L2, and their overlap at the same 25% retention. Green and red outlines identify units kept before and after merging, respectively. These cases illustrate spatial allocation rather than aggregate accuracy.

Efficiency and Scope

Efficiency is measured in vLLM 0.19 V1 on one A40 GPU. Across 75%, 50%, and 25% retention, RBM and Post-L2 throughput is 4.61/4.52, 5.39/5.54, and 6.91/6.89 requests/s. Five-repeat measurements remain within 3% with no consistent stage difference. Relative to no compression, RBM raises throughput by 12%, 31%, and 68%; at 25% retention, latency falls from 0.36 to 0.23 seconds. InternVL3 independently yields $1.8\text{--}2.5\times$ throughput at 25% retention. RBM does not reduce ViT computation before its scoring tap; its gain comes from shortening the sequence presented to the native merger and language model.

The byte-exact ranking attribution is established on Qwen3-VL. Qwen2.5-VL preserves kept-set identity but has an ordering confound, and InternVL3 offers accuracy-level replication and a bounded swap result. Models without an identifiable native merge unit are outside the method’s scope. FastV and RankBridge accuracy uses a different eager harness from the vLLM efficiency study. Several secondary FastV

and mechanism cells use $n = 200$, as marked, and only the TextVQA RankBridge gain is significant. Pixel caps differ across some model families, so absolute cross-model scores should not be compared. The current result supports a merger-aware machine-centric design rule for text-dense compression, not a universal ranking stage or selector.

Conclusion

Visual-token selection around a native merger is an operator-order problem. RBM resolves it by selecting complete native units on merger-input features and applying the unchanged model only to the survivors. Under equal token budgets, this training-free design preserves substantially more text-dense evidence than matched Post-L2 across three VLMs and two merger designs while delivering the expected inference speedup. A locked merger-aware rank prior also improves FastV-k3 on all four tested Qwen3-VL probes, indicating that pre-merger and query-conditioned ranks contribute complementary information. The mechanism results show why: the native merger changes saliency order and preferentially demotes text-stroke units. Compression methods for merger-equipped VLMs should therefore treat native topology and selection stage as first-class design constraints.

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